

Hidden Liquidity, Displayed Depth, and Execution Risk During a Bitcoin Perpetual Futures Liquidation Cascade

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Abstract

I study hidden liquidity in Bitcoin perpetual futures during a liquidation cascade, using a complete trade tape and fifty-level order book from the Bybit BTCUSDT contract over a thirty-nine-hour window containing a nine percent decline, with four additional stress events for robustness. The raw tape of 7.16 million sub-fills is aggregated into 14,704 parent market orders. For each parent order I compare the slippage implied by walking the visible book against realized slippage to detect hidden depth and estimate its size relative to displayed inside depth. Hidden liquidity is detected more often during the cascade than in calmer events, and the estimated hidden-depth ratio is larger; the cascade effect survives controls for order size, direction, and snapshot lag. Displayed depth is a biased and incomplete proxy for executable liquidity precisely when execution risk is greatest.

Keywords: market microstructure; perpetual futures; hidden liquidity; order book; liquidation cascade; execution risk

JEL classification: G12; G14; G15; C58

1. Introduction

A liquidation cascade is the moment a leveraged market is most likely to fail the traders who supply it. As forced sales push price through successive liquidation thresholds, the depth a liquidity provider believed it could rely on may evaporate, while latent depth that never appeared on the screen may quietly absorb the flow. For anyone supplying or consuming liquidity in crypto perpetual futures, the practical question is therefore not what the displayed order book shows, but how well that display predicts the cost of actually being filled when the market is breaking.

This paper addresses that question directly. Using a complete capture of the trade tape and the fifty-level limit order book for the Bybit BTCUSDT perpetual contract, I examine a thirty-nine-hour window from 2 June 2026 during which the contract fell from approximately 69,400 to 63,200, a decline of about nine percent, broadly consistent with the contemporaneous spot-market Bitcoin sell-off. The raw tape of 7.16 million sub-fills is aggregated into 14,704 parent market orders. For each parent order I compare the slippage implied by walking the displayed depth against the realized slippage of the fill. When the order executes at a better price than the visible book could have provided, the difference identifies hidden liquidity, and I estimate the size of that hidden depth relative to the displayed depth at the inside.

Three findings emerge and are consistent across all five events. First, realized slippage frequently falls short of the slippage predicted by the visible book, so displayed depth understates the liquidity actually available to absorb an order. Second, hidden liquidity is detected more often during the cascade than in the four calmer May 2026 events used for robustness, indicating that latent depth surfaces under stress rather than withdrawing. Third, the

estimated hidden-depth ratio is larger during the cascade, so the gap between displayed and true available liquidity widens precisely when execution risk is highest.

The contribution is to bring trade-level evidence on hidden liquidity to a stress episode in a crypto perpetual market, a setting where displayed depth is widely used as a risk proxy but rarely validated against realized execution. The detection and estimation approach draws on the hidden-liquidity literature discussed below, inferring latent depth from the discrepancy between the visible-book walk and the realized fill. The findings carry a direct message for digital-asset derivatives desks, complementing recent work in this journal on asset behaviour under stress (Mentes Usman et al., 2026): during forced deleveraging, the displayed book is an incomplete and biased proxy for executable liquidity, and execution decisions that rely on it will misstate the liquidity actually present.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on hidden liquidity, limit-order-book trading, and crypto-derivatives microstructure and states the contribution. Section 3 sets out the measurement framework, including a formal statement of the hidden-depth estimator. Section 4 describes the data and event sample. Section 5 reports the results, and Section 6 analyses the cascade by regime. Section 7 reports a pooled robustness regression and states the limitations, and Section 8 concludes.

2. Literature and contribution

The economics of limit order books rest on adverse selection. A resting limit order is filled by counterparties who may hold superior information, so the liquidity provider faces the classic problem articulated by Kyle (1985) and by Glosten and Milgrom (1985), in which the bid-ask spread compensates for the risk of trading against the informed. Easley and O'Hara (1987) link trade size to information, and Glosten (1994) analyses the viability of the electronic limit order

book itself. Empirical decompositions of the spread and of price impact follow Hasbrouck (1991) and Madhavan, Richardson, and Roomans (1997), while Biais, Hillion, and Spatt (1995) document the dynamics of supply and demand in an open order book. Sandås (2001) provides direct empirical evidence on adverse selection in a pure limit order market.

Hidden liquidity complicates this picture. Iceberg and hidden orders let traders display only a fraction of their size, so the visible book is an incomplete representation of available depth. Frey and Sandås (2017) show that when market participants detect iceberg orders they respond with matching market orders, that the price impact of larger executed hidden fractions is smaller in a manner consistent with liquidity rather than information, and that liquidity suppliers earn the highest surplus when undetected hidden liquidity is present. Esser and Mönch (2007) characterise the optimal use of hidden orders, and Boehmer, Saar, and Yu (2005) study how pre-trade transparency affects the book. A related strand develops estimators of hidden depth from publicly observable order-book dynamics, showing that inferred hidden liquidity improves price-impact prediction. The present paper sits in this tradition: it infers hidden depth from the discrepancy between the visible-book walk and the realized fill, a revealed-execution approach that requires no proprietary message data.

Crypto perpetual futures are now the dominant instrument in digital-asset trading, with perps accounting for the large majority of Bitcoin futures volume and open interest (Ackerer, Hugonnier, and Jermann, 2025). The funding-rate mechanism that anchors the perpetual to its index is central to their microstructure, and empirical work has begun to document price discovery and cross-venue structure in these markets (Alexander, Choi, Park, and Sohn, 2020; De Blasis and Webb, 2022). Liquidation cascades, in which forced sales trigger further liquidations in a self-reinforcing loop, are a recurring and consequential feature of

leveraged crypto markets, yet trade-level evidence on how liquidity provision behaves through such episodes remains scarce. This paper contributes such evidence, focusing on hidden liquidity as the channel through which displayed depth misstates execution risk during forced deleveraging.

3. Measurement framework

The unit of analysis is the parent market order. The raw tape records each sub-fill of a marketable order separately; sub-fills sharing a trade identifier and timestamp are aggregated into a single parent order with signed direction s (equal to $+1$ for buyer-initiated and -1 for seller-initiated orders), total size Q , and volume-weighted fill price VWAP. Aggregation to the parent order is the economically meaningful unit because a single marketable order that sweeps several price levels is reported by the exchange as multiple sub-fills; the slippage borne by the trader is the volume-weighted average over those sub-fills, not the slippage of any one of them, so the decomposition is performed at the parent level. For each parent order the contemporaneous order book is taken from the most recent snapshot preceding the order, subject to a maximum snapshot lag; orders whose nearest snapshot exceeds the lag tolerance, or for which the book is degenerate, are discarded.

Let m denote the pre-trade midpoint, the average of the best bid and best ask. Realized slippage is measured in trade-direction terms, so that a positive value denotes an adverse fill relative to the midpoint regardless of side,

$$dp_{real} = s \cdot (VWAP - m) ,$$

expressed in contract price units (USDT). The visible-book slippage is obtained by walking the displayed depth: filling size Q against the resting levels of the book yields a volume-weighted execution price $P_{walk}(Q)$, and, in the same trade-direction convention,

$$dp_walk = s \cdot (P_walk(Q) - m) .$$

If displayed depth is insufficient to fill Q , the walk is taken to the book's limit. Hidden liquidity is identified when the order fills better than the visible book could have provided, that is, when the slippage predicted by walking the displayed book exceeds the realized slippage by more than a one-tick tolerance ε :

$$dp_walk > dp_real + \varepsilon .$$

Because both quantities are expressed in trade-direction terms, the detection rule is symmetric across buyer- and seller-initiated orders, and the comparison is restricted to the sign-consistent region in which the realized fill lies on the expected side of the midpoint. The magnitude of hidden depth is then recovered as the quantity d that, when removed from the order before walking the residual against the visible book, reproduces the realized slippage:

$$\hat{d} = \arg \min_d |P_walk(Q - d) - m| - |dp_real| ,$$

solved by bisection. The following proposition establishes that this estimator is well defined and conservative on the domain over which detection occurs.

Proposition 1 (Existence and conservatism of the hidden-depth estimator). Let the visible book consist of discrete price levels $i \in \{1, \dots, N\}$ with non-negative depths v_i , so that the visible-book execution price $P_walk(q)$ is a right-continuous, non-decreasing step function of the filled quantity q . On the sign-consistent domain on which detection is defined — where the residual walk $P_walk(Q - d)$ remains on the expected side of the midpoint m — the implied visible slippage $S(d) = |P_walk(Q - d) - m|$ is a non-increasing step function of the hidden depth $d \in [0, Q]$. Consequently the objective $f(d) = |S(d) - |dp_real||$ attains its minimum on a closed interval $D = [d_min, d_max]$, and the estimator defined as the infimum

$$\hat{d} = \inf D$$

is unique and constitutes a conservative lower bound on the hidden depth consistent with the realized fill.

Intuitively, because visible-book walk slippage is weakly decreasing in inferred hidden depth over the sign-consistent domain, the estimator is well defined; where several hidden-depth values reproduce the same slippage on a step-function plateau, choosing the smallest makes the estimate conservative, so any reported increase in hidden liquidity during a cascade reflects the data rather than plateau width. The monotonicity holds on the sign-consistent domain to which the detection rule already confines the search, so the restriction is substantive rather than cosmetic. In implementation the bisection terminates when the bracketing interval narrows below a tolerance of 0.001 in depth units; relative to the inside depth V used to normalise the estimate this is smaller than the reported precision, so the choice of the infimum does not affect any figure reported below.

The headline statistic normalises recovered hidden depth by the visible depth at the inside, V :

$$\hat{\kappa} = \hat{d} / V,$$

a dimensionless ratio giving hidden depth as a fraction of displayed inside depth. The quantity $\hat{\kappa}$ is a hidden-depth ratio, not an estimate of adverse selection: it measures how much latent depth stood behind the displayed inside, and the language of adverse selection is reserved for interpretation. Because $\hat{d}$ and $\hat{\kappa}$ are defined only on orders for which hidden liquidity is detected, the $\hat{\kappa}$ series is conditional on detection; the number of identifying observations is reported for every event, and statistics on $\hat{\kappa}$ are computed over detected orders only. A regime classifier, computed at approximately one-minute cadence from quote volatility, a trade-intensity measure, and a smoothed spread statistic, tracks the intensity of the cascade for the analysis in Section 6.

4. Data and event sample

The data are captured directly from the public market-data interface of Bybit and consist of the complete trade tape and fifty-level order-book snapshots for the BTCUSDT perpetual contract, each record carrying both exchange and local-receipt timestamps at millisecond resolution. The headline event is a thirty-nine-hour window from 2 June 2026, 12:00 UTC to 4 June 2026, 03:00 UTC, during which the contract declined from approximately 69,400 to 63,200. The window contains 7,163,114 raw sub-fills, which aggregate into 14,704 parent market orders satisfying the snapshot-lag and book-walk criteria; these parent orders are the observations on which the decomposition is computed. Hidden liquidity is detected on 9.2 percent of them, yielding 1,347 orders with an estimated hidden-depth ratio. Trade intensity rises sharply during the break: the five-minute count of raw tape events peaks at 89,427 against a calmer-period level on the order of three thousand.

Four further stress events from May 2026, on the same contract and venue, provide robustness: events beginning 13 May (3,435 parent orders; 213 with a hidden-depth estimate), 15 May (4,667; 322), 16–18 May (10,806; 624), and 22 May (1,384; 84). Hidden-liquidity detection rates in these calmer events range from 5.8 to 6.9 percent, below the cascade, and the mean hidden-depth ratio is lower, approximately 0.18 to 0.19 against 0.25 in the cascade. The two larger May events corroborate the cascade most strongly; the two smaller events, with fewer identifying observations, are reported as supporting evidence and interpreted with appropriate caution. All summary statistics reported in this paper are computed from these captures; no figures are simulated or imputed.

The analysis concerns execution slippage relative to displayed depth, not the net profit and loss of any position. Trading fees and funding payments, which determine net economics but not the

relationship between displayed and realized execution prices, therefore do not enter the decomposition and are not modelled.

Table 1 reports the event summary.

Event	Parent orders	Hidden detections	Hidden detected (%)	Mean hidden-depth ratio	Median hidden-depth ratio	Median excess visible slippage (USDT)
June cascade	14,704	1,347	9.2	0.2501	0.1027	0.7739
13 May	3,435	213	6.2	0.1866	0.0824	0.6879
15 May	4,667	322	6.9	0.1756	0.0792	0.8994
16–18 May	10,806	624	5.8	0.1940	0.0824	0.8306
22 May	1,384	84	6.1	0.1912	0.0832	1.1346

Table 1. Event summary for the June cascade and four May stress events: parent orders, hidden-liquidity detections, detection rate, hidden-depth-ratio statistics over detected orders, and excess visible slippage. Excess visible slippage is the median of $|\text{dp_walk}| - |\text{dp_real}|$ in USDT over detected orders.

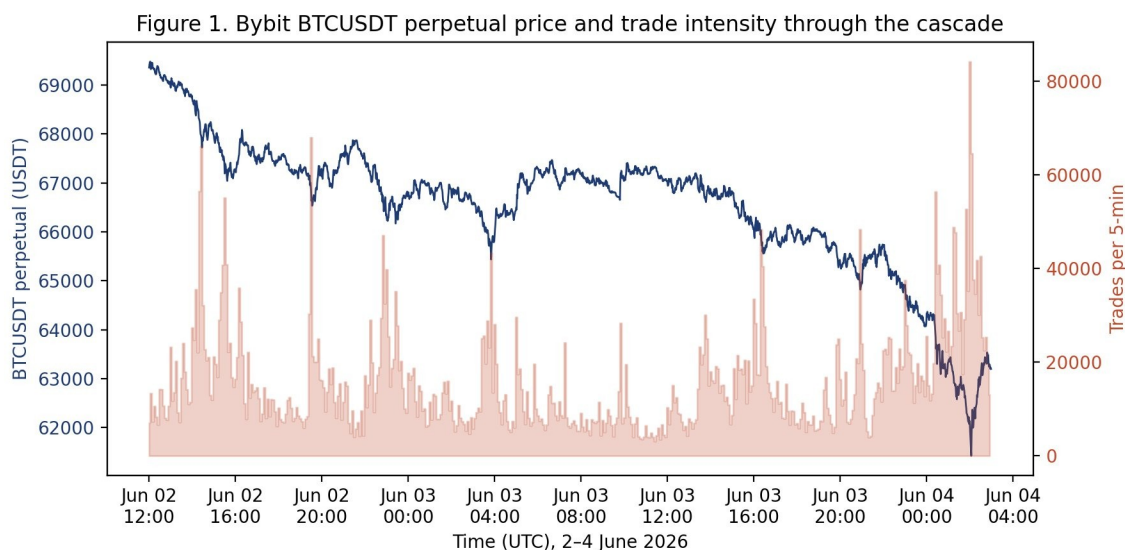


Figure 1. Bybit BTCUSDT perpetual price (one-minute last) and trade intensity (five-minute trade count) over the 2–4 June 2026 cascade window. The contract declines from approximately 69,400 to 63,200, with trade intensity peaking near the price trough.

5. Results

Across every event the visible book understates the liquidity available to absorb an order. The mean excess of visible-book slippage over realized slippage is positive throughout, and in the cascade the median realized-versus-visible gap is 0.77 USDT, rising to over one USDT in the smallest May event. The interpretation is that orders are routinely filled at better prices than displayed depth alone would predict, evidence that latent depth stands behind the screen and is consumed as orders arrive. Displayed depth is, in this sense, a biased and incomplete proxy for executable liquidity rather than a faithful measure of it.

Hidden-liquidity detection is more frequent during the cascade than in the calmer May events, at 9.2 percent against a range of 5.8 to 6.9 percent. Latent supply that the displayed book does not show is therefore more, not less, prevalent under stress, consistent with sophisticated participants withholding displayed size and supplying liquidity opportunistically as the cascade proceeds. The hidden-depth ratio moves in the same direction, with a mean of 0.25 in the cascade against 0.18 to 0.19 in the May events and a heavier upper tail, indicating that the quantity of hidden depth relative to displayed depth is greatest when the market is breaking.

Table 2 reports the cascade regime statistics.

Statistic	Calm-period level	Cascade peak
Quote volatility (mean σ_q)	—	0.2917
Smoothed spread statistic	≈ 0	0.8524
Structural parameter (κ^*)	0.50	0.93
Five-minute trade count	$\approx 3,000$	89,427

Table 2. Cascade regime statistics: quote volatility, the smoothed-spread statistic, the regime classifier's structural parameter, and the five-minute trade count, comparing the calm-period level with the cascade peak over the 2–4 June 2026 window.

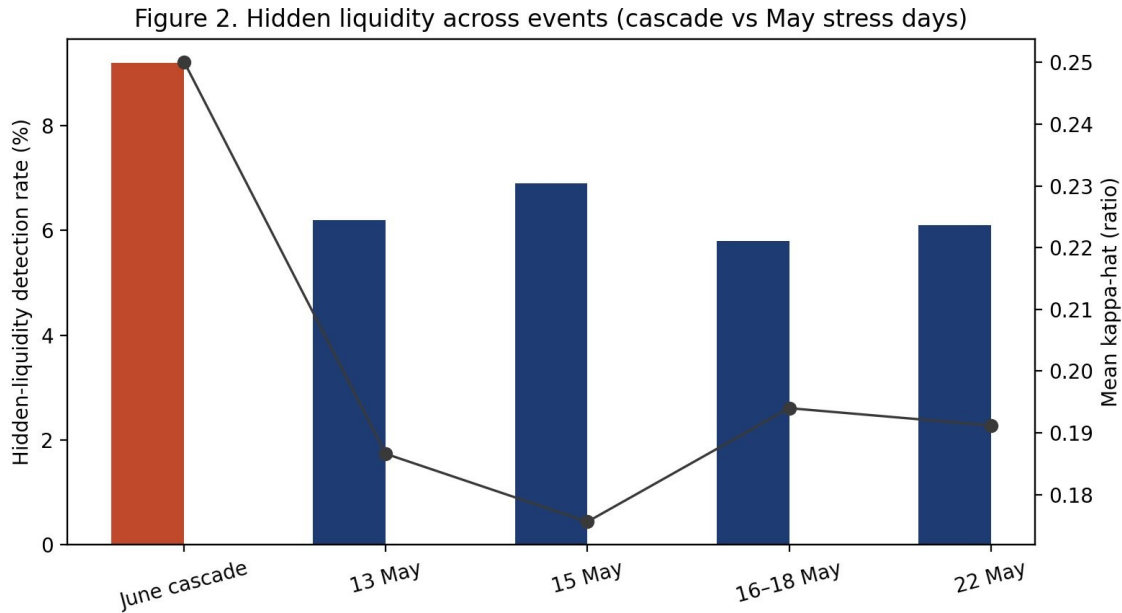


Figure 2. Hidden-liquidity detection rate (bars, left axis) and mean hidden-depth ratio (line, right axis) across the five events. The June cascade shows both a higher detection rate (9.2 percent) and a larger mean ratio (0.25) than the four May stress events.

6. Regime analysis

Tracking the cascade by the regime classifier locates the effect in the most intense phase of forced deleveraging. The smoothed spread statistic rises from a calm-period floor near zero toward a peak of 0.85, and the classifier's structural parameter rises from 0.50 to 0.93, while the five-minute trade count climbs from roughly three thousand to its peak of 89,427 before subsiding. Hidden-liquidity detection and the hidden-depth ratio are both elevated when these stress indicators are highest, so the widening gap between displayed and true available liquidity is concentrated in the most acute part of the cascade rather than spread evenly across the window.

A finer split of the cascade window around the intensity peak shows the within-window effect to be present but modest: the detection rate rises from 8.9 percent in the pre-peak phase to 10.5 percent at the peak, with the mean hidden-depth ratio rising correspondingly. The elevation

relative to the calmer May events is therefore the stronger and more robust comparison; the within-cascade gradient is consistent in direction but smaller in magnitude. The capture window ends close to the price trough, so the subsequent recovery is not observed, and within-cascade results are interpreted with that limitation in mind.

7. Robustness and limitations

A natural concern is that the higher hidden-liquidity detection rate in the cascade reflects nothing more than a different composition of order flow: cascade periods may contain larger orders, a different balance of buyers and sellers, or shorter snapshot lags, any of which could mechanically raise detection. To address this, I pool all five events into a single cross-section of 34,996 parent orders, of which 7.4 percent record a hidden-liquidity detection, and regress an indicator for detection on a cascade dummy (equal to one for the June window and zero for the four May events) together with the natural logarithm of order size, a sell indicator, and the snapshot lag in milliseconds. Both a linear probability model with heteroskedasticity-robust standard errors and a logit specification are estimated.

The controls enter as expected and are individually significant: larger orders are more likely to reveal hidden depth (a coefficient of 0.014 per log unit of size in the linear model, $z \approx 8.6$), seller-initiated orders slightly less likely (-0.009 , $z \approx -3.1$), and longer snapshot lags marginally less likely ($z \approx -4.2$). Crucially, the cascade dummy remains positive and strongly significant after these controls, at 0.027 in the linear probability model ($z \approx 9.3$, $p < 0.001$) and 0.396 in the logit ($z \approx 9.6$, $p < 0.001$). On the pooled base rate of 7.4 percent, the linear estimate implies that the cascade raises the probability of detecting hidden liquidity by roughly 2.7 percentage points beyond what order size, direction, and lag explain. The cascade effect is therefore not an artifact

of order-flow composition: part of the raw gap does operate through order size, but a distinct cascade-specific increase in hidden-liquidity detection survives controls.

Table 3 reports the pooled regression.

Regressor	LPM coef.	LPM z	Logit coef.	Logit z
Cascade	0.0273	9.27	0.3959	9.59
log(order size)	0.0140	8.58	0.1887	8.51
Sell	−0.0086	−3.06	−0.1268	−3.09
Snapshot lag (ms)	−0.00008	−4.17	−0.0012	−4.33
Constant	0.0679	19.82	−2.6389	−51.80

Table 3. Pooled robustness regression of a hidden-liquidity detection indicator on a cascade dummy and controls, over 34,996 parent orders from all five events (pooled detection rate 7.4 percent). The linear probability model (LPM) is estimated with heteroskedasticity-robust (HC1) standard errors; the logit is estimated by maximum likelihood. The cascade coefficient is positive and significant at the one percent level in both specifications.

Several limitations should nonetheless be stated plainly. First, the hidden-depth estimates are identified only on orders for which realized slippage falls short of the visible-book walk, so the $\hat{\kappa}$ series is conditional on detection and the two smallest May events rest on fewer than a few hundred identifying observations. Second, the analysis covers a single venue and a single contract; the magnitudes reported here need not generalise to other venues or instruments, and a cross-venue study is left to future work. Third, the detection rule depends on a snapshot-lag tolerance and a one-tick tolerance margin, and on the accuracy of the fifty-level book in representing depth at the inside; alternative thresholds and lag tolerances are natural sensitivity checks. Fourth, the decomposition measures realized execution against displayed depth and does not separately identify the informational versus inventory components of any residual impact, so the language of adverse selection is used here only as interpretation, not as a separately estimated parameter; the estimated quantity $\hat{\kappa}$ is a hidden-depth ratio.

8. Conclusion

Through a liquidation cascade in the Bybit Bitcoin perpetual, the displayed order book is an incomplete and biased guide to the liquidity actually available. Orders fill better than the visible book predicts, hidden liquidity is detected more often than in calmer periods, and the estimated hidden-depth ratio is larger, all concentrated in the stress phase of forced deleveraging. The practical implication for digital-asset derivatives desks is not the truism that crypto markets are risky, but the sharper point that displayed depth is an incomplete risk measure precisely when it is most relied upon: execution and liquidity-provision decisions calibrated to the visible book will misstate available liquidity during a cascade, in the direction of understating it.

Data availability statement

The trade and order-book data analysed in this study were captured by the author from the public market-data interface of the Bybit BTCUSDT perpetual contract. Derived datasets, the per-order decomposition output, and the estimation code supporting the findings are available from the author on reasonable request. Raw exchange data are subject to the venue's terms of use.

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Notes

This section is reserved for numbered endnotes in accordance with the journal's instructions. No substantive endnotes are required in the present version.